Zilionis 2019 NSCLC — analysis summary

RUN 5 OF 6

GO / NO-GO GATES

Gate 1 — integration & continuity → GO

All 6 samples and both histologies intermixed on ONE connected manifold (sample iLISI 3.4/6 at θ=10; 0 detached islands). Mixed + continuous — judged on the embedding, not a scalar.

Gate 2 — module geography → GO (non-branching)

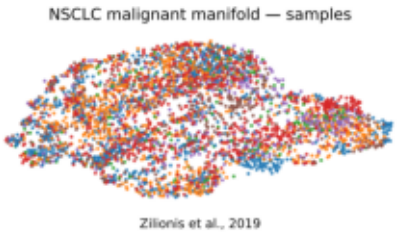
Shared developmental gradient: monotone Gast(origin, ρ=−0.59) → pSom(terminus, ρ=+0.76) axis; Crest an inverse off-axis readout. Present but collapsed onto one axis, not two competing arms.

Gate 3 — hub cross-repression → NA

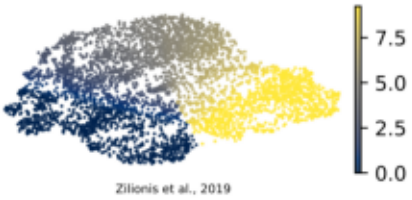
No Gast/pSom co-expression HUB (co-occurrence enrichment 0.07 smoothed / 0.70 raw, both <1) and no stable branch point (transcriptome PPT = 2 tips/0 forks across nodes × seeds). Non-branching → NA.

STEP FIGURES

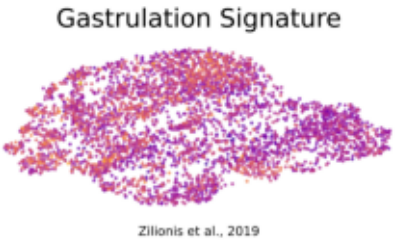
Manifold — samples



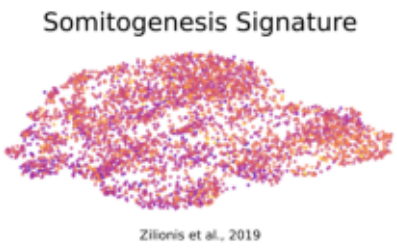
Pseudotime



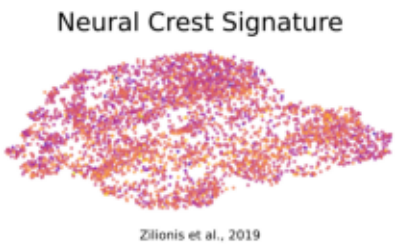
Gastrulation



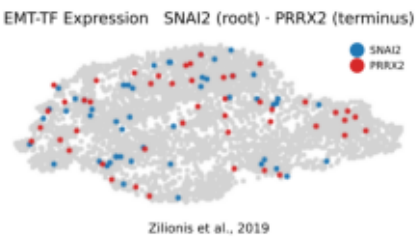
Somitogenesis



Neural Crest



EMT-TF expression



Dataset.
NSCLC (Zilionis et al., 2019), inDrop — 3,920 malignant cells (6/7 samples; mixed LUSC + LUAD histology). Geometry: LINEAR (non-branching).

Verdict.
The embryonic pattern reproduces on a single collapsed axis — a gastrulation-high origin resolving to a somitogenesis-high terminus, with neural crest as an inverse off-axis gradient (flagged: non-terminal). Not the Puram hub-and-spoke bifurcation. Gate 1 GO / Gate 2 GO / Gate 3 NA. EMT-TFs SNAI2 (early/root) ↔ PRRX2 (late/terminus); weak, root-loaded.

Parameter choices

Threshold	≥131 malignant (geom-mean across 232→75 count cliff, ratio 3.09) → drops p2 (75) → 3,920 cells	**Overdispersed genes**	248 (scFates GAM); PCA 50 PC, n_pcs=20 (~50% variance)
Integration	Harmony θ=10 (iLISI plateau 1.4→3.4/6); n_neighbors=30; UMAP spread=1.3; 0 islands	**Trajectory**	scFates CURVE (ElPiGraph) N=20 λ=0.3 μ=100; 2 tips/0 forks, fold-free (SA 2.45); corr(pt,DC1)=−0.85
Root	Gast-high tip (Crest is late, never the root)	**EMT-TFs**	SNAI2 (early/root) ↔ PRRX2 (late/terminus); weak, root-loaded